

M. Sc. Physics (Semester-II) 2026

Quantum Mechanics II – Tutorial 8

Tutorial exercises

1. An electron makes a transition from an initial discrete state $|i\rangle$ to continuum of states $|f\rangle$ with energies E_f under a constant perturbation. The matrix element is given as $|\langle f|V|i\rangle|^2 = A$ and the density of state is $\rho(E) = \sqrt{E}$. Write down the expression for the transition rate and how does the transition rate depend on energy?
2. Consider a particle of mass m in a potential $U(x)$ such that $U(x) < 0$ everywhere and vanishes for as $x \rightarrow \pm\infty$. Let $\psi_0 = \frac{a^{1/2}}{\pi^{1/4}} e^{-\frac{a^2 x^2}{2}}$ be the ground state wave function with energy $E_0 < 0$. Let the system be subject to a harmonic potential $V \cos \omega t$.
 - (a) What is the minimum value of ω for which we can apply the Fermi Golden Rule for transitions to continuum energy states?
 - (b) Assume the density $\rho(E)$ of free particle states is independent of energy. Let $\omega = -2E_0/\hbar$. Is the Fermi Golden Rule applicable? If so, apply it to calculate the transition rate to free particle states with energy $-E_0$.

Home exercise

3. Griffith (3rd edition) example 11.2 .